

Village Growth Intelligence: Top 100 Villages

This project identifies villages that show strong recent growth signals using village boundaries, satellite night-time lights, and built-area change. The final output is a ranked CSV of the top villages for further review.

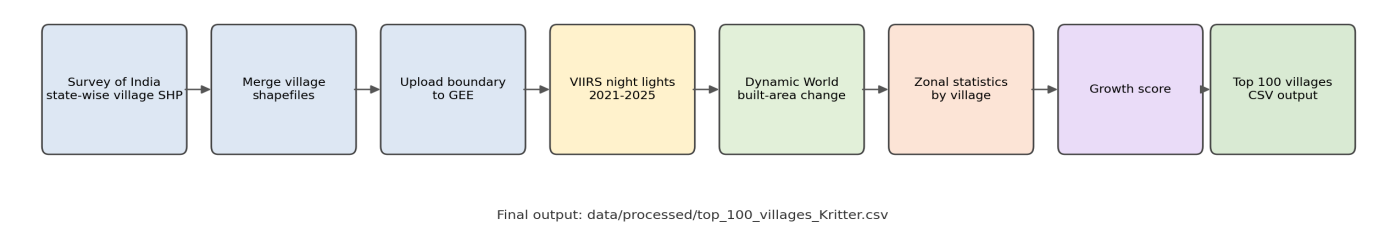
Why This Dataset Is Useful

For industry, this type of dataset helps find places where economic activity is becoming more visible. It can support market expansion, rural distribution planning, infrastructure prioritisation, financial inclusion, telecom planning, and early-stage location screening. Instead of relying only on old administrative statistics, it uses repeatable satellite signals that can be updated over time.

Methodology

I downloaded state-wise village boundary shapefiles from the Survey of India village boundary database, merged the readable state files into one combined village shapefile, uploaded it to Google Earth Engine, and calculated village-level growth signals.

Village Growth Intelligence Workflow



Input	Use in analysis
Survey of India village boundaries	Village polygons for zonal statistics
VIIRS night-time lights, 2021-2025	Proxy for visible economic activity
Dynamic World built-area signal	Supporting indicator for construction/settlement growth

Finding and Interpretation

The top-ranked villages are places where night-time brightness increased strongly compared with the 2021 baseline. In simple terms, these locations became more visibly active at night, which often reflects more lighting, settlement activity, shops, roads, services, or local commercial movement. These results should be treated as a screening list: they point to villages worth deeper ground validation rather than proving income growth by themselves.

Final Output

Final CSV: `data/processed/top_100_villages_Kritter.csv`. Code: `gee/village_growth_feature_extraction.js`. Methodology note: `docs/methodology.md`.

Main limitation: some source shapefiles were missing or damaged, so the final boundary layer reflects the readable village shapefiles available during processing.